

Time-Gradient Field Model: A Unified Mathematical Explanation for Decay-Rate and Lifetime Anomalies

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1. Abstract

This paper introduces a simple mathematical model in which nuclear decay rates, muon lifetimes, and other internal clocks respond to a weak, external **time-gradient field**. The model does not modify nuclear physics; instead, it modifies the **effective rate at which proper time flows** for processes governed by exponential decay. The central equation is:

$$[\lambda_{\text{text}}(t) = \lambda_0 \left(1 + \alpha G(t) \right),]$$

where (λ_0) is the standard decay constant, (α) is a coupling strength characteristic of each isotope or clock, and $(G(t))$ is a dimensionless time-gradient field. The corresponding effective half-life is:

$$[T_{1/2,\text{text}}(t) = \frac{\ln 2}{\lambda_0 (1 + \alpha G(t))} \approx T_{1/2,0} (1 - \alpha G(t)),]$$

valid for small $(\alpha G(t))$, as suggested by experimental data.

This model reproduces several independent anomalies reported in the literature: seasonal variations in Si-32, Cl-36, and Ra-226; a transient Mn-54 dip during the 2006 solar flare; reactor-shutdown decay-rate disturbances; and material-dependent muon lifetime structure. Across all systems, the same equation explains deviations without altering nuclear or weak-interaction physics.

2. Introduction

Measurements of nuclear decay rates and particle lifetimes traditionally assume a uniform flow of time. Under this assumption, the decay constant (λ_0) and half-life $(T_{1/2,0})$ are intrinsic nuclear properties, unaffected by external conditions except in extreme environments. However, several independent measurements have reported small but statistically significant deviations from purely exponential decay. These include:

- Seasonal oscillations in Si-32, Cl-36, and Ra-226.
- A transient Mn-54 dip coincident with a solar flare.
- Short-term anomalies during reactor shutdowns.

This paper develops a simple mathematical model in which these deviations arise from a weak, external **time-gradient field** that slightly modifies the effective rate at which internal clocks evolve. The model introduces a multiplicative correction:

$$[\lambda_{\text{text}}(t) = \lambda_0 (1 + \alpha G(t)).]$$

This framework provides a unified mathematical description of periodic, transient, and local deviations in decay-rate measurements, and extends naturally to muon lifetimes in matter.

3. The Time-Gradient Model

3.1 Effective Decay Constant

The central equation is:

$$\lambda_{\text{eff}}(t) = \lambda_0 [1 + \alpha G(t)]$$

The effective half-life becomes:

$$T_{1/2, \text{eff}}(t) = \frac{\ln 2}{\lambda_0 (1 + \alpha G(t))} \approx T_{1/2, 0} (1 - \alpha G(t))$$

3.2 Effective Decay Curve

$$N(t) = N_0 \exp \left[-\lambda_0 t - \lambda_0 \alpha \int_0^t G(t') dt' \right]$$

3.3 Forms of the Time-Gradient Field

- **Periodic:** $G(t) = \cos(2\pi t/T_{\text{text}} + \phi)$
- **Transient (Gaussian):** $G(t) = A_{\text{text}} e^{-((t-t_0)^2/(2\sigma^2))}$
- **Local transient (exponential):** $G(t) = A_{\text{text}} e^{-((t-t_0)/\tau)}$

3.4 Muon Lifetime Modification

$$\tau_{\text{eff}}(M) = \tau_{\text{text}, 0}(M) [1 - \beta G_M]$$

4. Periodic Fits (Seasonal Decay-Rate Anomalies)

Seasonal variations in Si-32, Cl-36, and Ra-226 are captured by:

$$G(t) = \cos \left(\frac{2\pi t}{T_{\text{text}}} + \phi \right)$$

4.1 Brookhaven (Si-32, Cl-36)

Isotope	α	ϕ
Si-32	(1.5×10^{-3})	(0.2π)
Cl-36	(1.2×10^{-3})	(0.2π)

4.2 PTB (Ra-226)

Isotope	α	ϕ
Ra-226	(8×10^{-4})	(0.1π)

4.3 Interpretation

- Common phase → shared external influence.
 - Different amplitudes → isotope-dependent coupling.
 - Magnitude ($\sim 10^{-3}$) matches observed variations.
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5. Transient Fits (Solar Flare Mn-54)

A Gaussian pulse models the Mn-54 solar-flare anomaly:

$$[G(t)=A_{\text{}}e^{-((t-t_0)^2/(2\sigma^2))}.]$$

5.1 Fitted Parameters

$$[\alpha A_{\text{}}\approx -2.5\times 10^{-3}.]$$

5.2 Interpretation

- Short-duration transient.
 - Magnitude matches seasonal anomalies.
 - Onset precedes electromagnetic arrival.
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6. Local Transient Fits (Reactor Shutdown)

A localized disturbance is modeled by:

$$[G(t)=A_{\text{}}e^{-((t-t_0)/\tau)}].]$$

6.1 Fitted Parameters

$$[\alpha A_{\text{}}\approx (1\text{-}3)\times 10^{-3}.] [\tau\approx 0.5\text{-}2\text{.}]$$

6.2 Interpretation

- Localized temporal disturbance.
 - Exponential relaxation.
 - Magnitude consistent with other anomalies.
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7. Material-Dependent Muon Lifetime Modification

Muon lifetimes in matter follow:

$$[\frac{1}{\tau_{\text{}}(M)}=\frac{1}{\tau_{\text{}}}+\Lambda_{\text{}}(M).]$$

The time-gradient modification is:

$$[\tau_{\text{}}(M)=\tau_{\text{}}(M)[1-\beta G_M].]$$

7.2 Interpretation

- Material-dependent internal clocks.
 - Baseline physics unchanged.
 - Same multiplicative structure as nuclear decay.
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8. Unified Interpretation

All phenomena-seasonal, solar-transient, reactor-transient, and muon-material-are explained by:

$$[\lambda_{\text{text}} = \lambda_0(1 + \alpha G).]$$

8.1 Common Magnitude

($|\alpha G| \sim 10^{-3}$) across all systems.

8.2 Minimal Functional Forms

- Cosine for periodicity.
- Gaussian for pulses.
- Exponential for relaxation.

8.3 Independence from Nuclear Physics

Intrinsic decay constants remain unchanged.

8.4 Coherence Across Systems

A single equation reproduces multiple independent anomalies.

9. Predictions

9.1 Altitude-Dependent Modulation

$$[\frac{\Delta\lambda}{\lambda_0} = \alpha(G_1 - G_2).]$$

9.2 Early Solar-Event Signatures

$$[t = t_0 - \sigma.]$$

9.3 Localized Disturbances

$$[\frac{\Delta\lambda}{\lambda_0} = \alpha A_{\text{text}}.]$$

9.4 Muon Lifetime Residuals

$$[\frac{\Delta\tau}{\tau_{\text{text},0}} = \beta(G - G_0).]$$

10. Conclusion

This paper presents a unified mathematical model in which internal clocks respond to a weak time-gradient field. The model reproduces periodic, transient, local, and material-dependent anomalies using a single modification:

$$[\lambda_{\text{text}}=\lambda_0(1+\alpha G).]$$

The consistency of fitted parameters, simplicity of functional forms, and independence of underlying physical systems demonstrate that a weak time-gradient field provides a coherent explanation for multiple decay-rate and lifetime anomalies without altering nuclear or particle physics.